Solutions and teaching notes, chapter 4

Teacher use. The following answers concern the tasks not solved in the lesson. First have pupils make their approach explicit; provide graduated help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. $\frac{8-2}{3-1} = 3$. *Action:* Represent horizontal and vertical increments.

2. $4h + h^2$. *Action:* Expand before simplifying.

3. $4 + h$. *Action:* Factor out h in the numerator before dividing.

4. 0 and 0,1. *Action:* Check by squaring.

5. 3 and 2. *Action:* Represent the distance from 0.

6. $y = 2x + 3$. *Action:* Substitute the coordinates to find the y-intercept.

Investigation

The slopes are 2 + h ; limit 2. The line passes through (1; 1), so $y = 2x - 1$. The drawing motivates, the calculation justifies.

Direct applications

1. $f(2+h) - f(2) = 7h + h^2$; rate 7 + h for $h \neq 0$.

2. The rate is 4 + 3 h, so the derivative at this point is 4.

3. $y = 5 - 2(x-3) = -2x$ + 11; approximation 5 $- 2(0{,}04) = 4{,}92$.

4. The rate at 2 is $\frac{|h|}{h}$, with the two opposite one-sided limits. For the square root at 0, the rate $\frac{1}{\sqrt{h}}$ for $h > 0$ exceeds every bound as h approaches 0: no real derivative.

Exercises: methods and answers

Exercise 4.1 - Two secants. Slopes 3 and 2.1; derivative 2. The second is closer to the tangent slope; here it is an exact calculation of a rate.

Exercise 4.2 - A tangent. Rate 4 $a+2h = -4+2h$, therefore $f\prime(-1) = -4$. $f(-1)$ = 3; tangent $y = 3 - 4(x+1) = -4x - 1$.

Exercise 4.3 - Marginal cost. $C(10) = 400$, $C(11)$ = 441, an increase of 41 for one unit. The rate around 10 is 40 + h, so the derivative is 40: approximation of the increment for a small change, not an equality for a whole unit.

Exercise 4.4 - Reading a tangent. On the figure in the lesson: points (1;0) and (2;4), slope 4. In the new situation, $f(2)$ = 3, slope $-\frac{1}{2}$. Approximation 3 $- 0{,}5 \times 0{,}1 = 2{,}95$. The y-coordinate of B is not necessarily $f(4)$.

Exercise 4.5 - Checking an approximation. $1+2(-0{,}02)=0{,}96$. Exact 0 ,9604; exact minus approximate 0 ,0004 >0. The term omitted is h^2 : an underestimate.

Exercise 4.6 - Tangent and intersection. The tangent has equation $y=1+3(x-1)=3x-2$. Expanding the product does give x^3-3x+2. The x-coordinates of the intersection points are therefore 1 and -2, by the zero-product rule. The points are (1; 1) and ($-2;-8$). There are two common points: this is enough to refute the general statement.

Exercise 4.7 - Numerical secants. List of rates 2+ h : 2.1; 2.01; 2.001 and 1.9; 1.99; 1.999. The algebraic calculation gives the real number 2. For very small h, floating-point calculation may lose precision.

Exercise 4.8 - Delayed review. For $h\neq 0,-1$, the rate is $-\dfrac{1}{1+h}$; it tends to -1. Approximation mation $1-0{,}02=0{,}98$. The exact value $\dfrac{50}{51}$ allows a check.

CQFD, exercise 17 p. 127, all questions. The x-coordinates of A, B and C are 0, 1 and 2. The tangents give, respectively, the slopes 2, 0 and -2. Thus $f\prime(0)=2,\ f\prime(1)=0,\ f\prime(2)=-2$. The axis units must be read from the grid.

CQFD, exercise 19 p. 127, questions 1 and 2. Point A is read as (1; 4), so $f(1)=4$. The tangent passes through (0; 1) and (1; 4): slope 3. Thus $f\prime(1) = 3$ and the tangent has equation $y=4+3(x-1)=3x+1$.

MCQ: answers and reasons

MCQ 1.

1. B. $\dfrac{4-1}{2-1}=3$.

2. A. $4+2(x-1)$.

3. D. It is a value at one point.

4. C. Rate 6 + h.

5. B. Affine tangent function.

MCQ 2.

1. C. Rate $\dfrac{1}{\sqrt{h}}$ not finite.

2. A. $|h|=-h$.

3. D. Constant y-coordinate.

4. B. A change of position divided by a change of time is expressed in m s $^{-1}$.

5. C. Exact equality for an affine function, approximation in general.

True or false

1. False: x^3 and the tangent 3 $x-2$ at 1 also meet at -2.

2. False: the rates on either side are -1 and 1.

3. True: the rate equals m for every nonzero h.

Differentiated teaching. If a pupil is stuck on a technical point, suggest only the first step (domain, factor or reference sample space), then remove the help. For an independent student, ask for a second method and the handling of a boundary case. Orally, allow an initial rephrasing in the language the pupil is fluent in, then require a precise mathematical sentence in French.